

BRIEF INSIGHTS and QUERY RESULTS OF SQL SCRIPT

- **Load Dataset into SQL Database** - This step creates the table structure and prepares the database for storing sales records.

Output -

	row_id integer	order_id character varying (25)	order_date character varying (15)	ship_date character varying (15)	ship_mode character varying (30)	customer_id character varying (15)	customer_name character varying (100)
1	1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute
2	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute
3	3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff
4	4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell
5	5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell
6	6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman
7	7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman
8	8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman
9	9	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman
10	10	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman
11	11	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman
12	12	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman
13	13	CA-2017-114412	4/15/2017	4/20/2017	Standard Class	AA-10480	Andrew Allen
14	14	CA-2016-161389	12/5/2016	12/10/2016	Standard Class	IM-15070	Irene Maddox
15	15	US-2015-118983	11/22/2015	11/26/2015	Standard Class	HP-14815	Harold Pawlan
16	16	US-2015-118983	11/22/2015	11/26/2015	Standard Class	HP-14815	Harold Pawlan
17	17	CA-2014-105893	11/11/2014	11/18/2014	Standard Class	PK-19075	Pete Kriz
18	18	CA-2014-167164	5/13/2014	5/15/2014	Second Class	AG-10270	Alejandro Grove
19	19	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925	Zuschuss Donatelli
Total rows: 9994		Query complete 00:00:00.515			CRLF Ln 11, Col 30		

- **Verify All Tables** - This query confirms all four tables are loaded correctly by displaying the row count of each table.

Output -

Showing rows: 1 to 4





Page No:

1

	table_name text	rows bigint
1	superstore_r...	9994
2	customers	2501
3	products	1894
4	orders	9994

Total rows: 4

Query complete 00:00:00.130

- **Orders Where Sales Are Greater Than Average Sales** - This query uses a subquery to calculate the average sales value and returns only those orders that exceed it.

Output -

Data Output Messages Notifications				
	order_id character varying (25)	customer_name character varying (100)	order_sales numeric	avg_sales numeric
1	CA-2014-145317	Sean Miller	22638.48	229.86
2	CA-2014-145317	Sean Miller	22638.48	229.86
3	CA-2014-145317	Sean Miller	22638.48	229.86
4	CA-2014-145317	Sean Miller	22638.48	229.86
5	CA-2016-118689	Tamara Chand	17499.95	229.86
6	CA-2016-118689	Tamara Chand	17499.95	229.86
7	CA-2016-118689	Tamara Chand	17499.95	229.86
8	CA-2016-118689	Tamara Chand	17499.95	229.86
9	CA-2017-140151	Raymond Buch	13999.96	229.86
10	CA-2017-140151	Raymond Buch	13999.96	229.86
11	CA-2017-140151	Raymond Buch	13999.96	229.86
12	CA-2017-140151	Raymond Buch	13999.96	229.86
13	CA-2017-127180	Tom Ashbrook	11199.97	229.86
14	CA-2017-127180	Tom Ashbrook	11199.97	229.86
15	CA-2017-166709	Hunter Lopez	10499.97	229.86
16	CA-2017-166709	Hunter Lopez	10499.97	229.86
17	CA-2017-166709	Hunter Lopez	10499.97	229.86
18	CA-2016-117121	Adrian Barton	9892.74	229.86
Total rows: 7945		Query complete 00:00:00.250		

- **Highest Sales Order for Each Customer** - A correlated subquery finds each customer's single highest value order from the dataset.

Output -

Showing rows: 1 to 1000 Page No: 1 of 3			
	customer_name character varying (100)	order_id character varying (25)	highest_order_sales numeric
1	Sean Miller	CA-2014-145317	22638.48
2	Sean Miller	CA-2014-145317	22638.48
3	Sean Miller	CA-2014-145317	22638.48
4	Sean Miller	CA-2014-145317	22638.48
5	Tamara Chand	CA-2016-118689	17499.95
6	Tamara Chand	CA-2016-118689	17499.95
7	Tamara Chand	CA-2016-118689	17499.95
8	Tamara Chand	CA-2016-118689	17499.95
9	Raymond Buch	CA-2017-140151	13999.96
10	Raymond Buch	CA-2017-140151	13999.96
11	Raymond Buch	CA-2017-140151	13999.96
12	Raymond Buch	CA-2017-140151	13999.96
13	Tom Ashbrook	CA-2017-127180	11199.97
14	Tom Ashbrook	CA-2017-127180	11199.97
15	Hunter Lopez	CA-2017-166709	10499.97
16	Hunter Lopez	CA-2017-166709	10499.97
Total rows: 2508		Query complete 00:00:19.924 CRLF Ln 12, Col 35	

- **Total Sales per Customer** - A CTE aggregates the total sales and order count for every customer in the dataset.

Output -

Showing rows: 1 to 1000					
	customer_name character varying (100)	segment character varying (30)	region character varying (20)	total_sales numeric	total_orders bigint
1	Sean Miller	Home Office	East	25043.07	5
2	Sean Miller	Home Office	Central	25043.07	5
3	Sean Miller	Home Office	South	25043.07	5
4	Sean Miller	Home Office	West	25043.07	5
5	Tamara Chand	Corporate	West	19052.22	5
6	Tamara Chand	Corporate	Central	19052.22	5
7	Tamara Chand	Corporate	South	19052.22	5
8	Tamara Chand	Corporate	East	19052.22	5
9	Raymond Buch	Consumer	Central	15117.35	6
10	Raymond Buch	Consumer	South	15117.35	6
11	Raymond Buch	Consumer	East	15117.35	6
12	Raymond Buch	Consumer	West	15117.35	6
13	Tom Ashbrook	Home Office	Central	14595.62	4
14	Tom Ashbrook	Home Office	East	14595.62	4
15	Adrian Barton	Consumer	Central	14473.57	10
16	Adrian Barton	Consumer	South	14473.57	10
17	Adrian Barton	Consumer	West	14473.57	10
18	Adrian Barton	Consumer	East	14473.57	10
19	Ken Lonsdale	Consumer	East	14175.23	12
Total rows: 2501		Query complete 00:00:00.221			

- **Customers Whose Total Sales Are Above Average** - A CTE computes total sales per customer, and a subquery filters only those who exceed the overall average.

Output -

Showing rows: 1 to 1000					
	customer_name character varying (100)	segment character varying (30)	region character varying (20)	total_sales numeric	avg_benchmark numeric
1	Sean Miller	Home Office	South	25043.07	2896.85
2	Sean Miller	Home Office	West	25043.07	2896.85
3	Sean Miller	Home Office	Central	25043.07	2896.85
4	Sean Miller	Home Office	East	25043.07	2896.85
5	Tamara Chand	Corporate	West	19052.22	2896.85
6	Tamara Chand	Corporate	Central	19052.22	2896.85
7	Tamara Chand	Corporate	East	19052.22	2896.85
8	Tamara Chand	Corporate	South	19052.22	2896.85
9	Raymond Buch	Consumer	Central	15117.35	2896.85
10	Raymond Buch	Consumer	South	15117.35	2896.85
11	Raymond Buch	Consumer	East	15117.35	2896.85
12	Raymond Buch	Consumer	West	15117.35	2896.85
13	Tom Ashbrook	Home Office	Central	14595.62	2896.85
14	Tom Ashbrook	Home Office	East	14595.62	2896.85
15	Adrian Barton	Consumer	East	14473.57	2896.85
16	Adrian Barton	Consumer	West	14473.57	2896.85
17	Adrian Barton	Consumer	Central	14473.57	2896.85
18	Adrian Barton	Consumer	South	14473.57	2896.85
19	Ken Lonsdale	Consumer	East	14175.23	2896.85
Total rows: 1014		Query complete 00:00:00.191			

- **Rank All Customers by Total Sales** - The RANK() window function assigns a sales rank to every customer, with rank 1 being the highest spender.

Output -

Data Output Messages Notifications					
Showing rows: 1 to 1000					
	sales_rank bigint	customer_name character varying (100)	segment character varying (30)	region character varying (20)	total_sales numeric
1	1	Sean Miller	Home Office	East	25043.07
2	1	Sean Miller	Home Office	Central	25043.07
3	1	Sean Miller	Home Office	South	25043.07
4	1	Sean Miller	Home Office	West	25043.07
5	5	Tamara Chand	Corporate	West	19052.22
6	5	Tamara Chand	Corporate	Central	19052.22
7	5	Tamara Chand	Corporate	South	19052.22
8	5	Tamara Chand	Corporate	East	19052.22
9	9	Raymond Buch	Consumer	Central	15117.35
10	9	Raymond Buch	Consumer	South	15117.35
11	9	Raymond Buch	Consumer	East	15117.35
12	9	Raymond Buch	Consumer	West	15117.35
13	13	Tom Ashbrook	Home Office	Central	14595.62
14	13	Tom Ashbrook	Home Office	East	14595.62
15	15	Adrian Barton	Consumer	Central	14473.57
16	15	Adrian Barton	Consumer	South	14473.57
17	15	Adrian Barton	Consumer	West	14473.57
18	15	Adrian Barton	Consumer	East	14473.57
19	19	Ken Lonsdale	Consumer	East	14175.23
Total rows: 2501		Query complete 00:00:00.179			

- **Row Numbers for Each Order Within a Customer** - ROW_NUMBER() with PARTITION BY restarts the count for each customer, numbering their orders from highest to lowest sale.

Output -

Data Output Messages Notifications					
Showing rows: 1 to 1000 Page No: 1					
	customer_name character varying (100)	order_id character varying (25)	order_date character varying (15)	sales numeric	order_rank_within_customer bigint
1	Aaron Bergman	CA-2016-140935	11/10/2016	341.96	1
2	Aaron Bergman	CA-2016-140935	11/10/2016	341.96	2
3	Aaron Bergman	CA-2014-156587	3/7/2014	242.94	3
4	Aaron Bergman	CA-2014-156587	3/7/2014	242.94	4
5	Aaron Bergman	CA-2016-140935	11/10/2016	221.98	5
6	Aaron Bergman	CA-2016-140935	11/10/2016	221.98	6
7	Aaron Bergman	CA-2014-156587	3/7/2014	48.71	7
8	Aaron Bergman	CA-2014-156587	3/7/2014	48.71	8
9	Aaron Bergman	CA-2014-156587	3/7/2014	17.94	9
10	Aaron Bergman	CA-2014-156587	3/7/2014	17.94	10
11	Aaron Bergman	CA-2014-152905	2/18/2014	12.62	11
12	Aaron Bergman	CA-2014-152905	2/18/2014	12.62	12
13	Aaron Hawkins	CA-2015-130113	12/27/2015	668.16	1
14	Aaron Hawkins	CA-2015-130113	12/27/2015	668.16	2
15	Aaron Hawkins	CA-2015-130113	12/27/2015	668.16	3
16	Aaron Hawkins	CA-2015-130113	12/27/2015	323.10	4
17	Aaron Hawkins	CA-2015-130113	12/27/2015	323.10	5
18	Aaron Hawkins	CA-2015-130113	12/27/2015	323.10	6
19	Aaron Hawkins	CA-2014-113768	5/13/2014	279.46	7
Total rows: 33500 Query complete 00:00:00.453					

- **Top 3 Customers by Total Sales** - RANK() is applied and filtered to return only the top 3 customers by total sales.

Output -

Data Output Messages Notifications					
Showing rows: 1 to 4 Page					
	customer_name character varying (100)	segment character varying (30)	region character varying (20)	total_sales numeric	sales_rank bigint
1	Sean Miller	Home Office	East	25043.07	1
2	Sean Miller	Home Office	Central	25043.07	1
3	Sean Miller	Home Office	South	25043.07	1
4	Sean Miller	Home Office	West	25043.07	1

- **Final Combined Query (JOIN + CTE + Window Function)** - This query brings everything together — it uses a CTE to aggregate sales, a JOIN to fetch customer details, and a Window Function to assign ranks. Output shows Customer Name, Total Sales, Total Profit, Total Orders and Rank.

Output -

Data Output Messages Notifications

Showing rows: 1 to 1000 Page No: 1 of 3

	Rank bigint	Customer Name character varying (100)	Segment character varying (30)	Region character varying (20)	Total Sales (\$) numeric	Total Profit (\$) numeric	Total Orders bigint
1	1	Sean Miller	Home Office	East	25043.07	-1980.75	5
2	1	Sean Miller	Home Office	Central	25043.07	-1980.75	5
3	1	Sean Miller	Home Office	South	25043.07	-1980.75	5
4	1	Sean Miller	Home Office	West	25043.07	-1980.75	5
5	5	Tamara Chand	Corporate	West	19052.22	8981.32	5
6	5	Tamara Chand	Corporate	Central	19052.22	8981.32	5
7	5	Tamara Chand	Corporate	South	19052.22	8981.32	5
8	5	Tamara Chand	Corporate	East	19052.22	8981.32	5
9	9	Raymond Buch	Consumer	Central	15117.35	6976.09	6
10	9	Raymond Buch	Consumer	South	15117.35	6976.09	6
11	9	Raymond Buch	Consumer	East	15117.35	6976.09	6
12	9	Raymond Buch	Consumer	West	15117.35	6976.09	6
13	13	Tom Ashbrook	Home Office	Central	14595.62	4703.80	4
14	13	Tom Ashbrook	Home Office	East	14595.62	4703.80	4
15	15	Adrian Barton	Consumer	Central	14473.57	5444.81	10
16	15	Adrian Barton	Consumer	South	14473.57	5444.81	10
17	15	Adrian Barton	Consumer	West	14473.57	5444.81	10
18	15	Adrian Barton	Consumer	East	14473.57	5444.81	10
19	19	Ken Lonsdale	Consumer	East	14175.23	806.84	12

Total rows: 2501 Query complete 00:00:00.337 CRLF

Mini Project: Customer Sales Insights

- **Top 5 Customers** - Identifies the five highest spending customers by total sales using RANK() and ORDER BY.

Output -

Data Output Messages Notifications

Showing rows: 1 to 5

	rank bigint	customer_name character varying (100)	segment character varying (30)	region character varying (20)	total_sales numeric
1	1	Sean Miller	Home Office	East	25043.07
2	1	Sean Miller	Home Office	Central	25043.07
3	1	Sean Miller	Home Office	South	25043.07
4	1	Sean Miller	Home Office	West	25043.07
5	5	Tamara Chand	Corporate	West	19052.22

- **Bottom 5 Customers** - Identifies the five lowest spending customers by total sales to highlight low engagement accounts.

Output -

Data Output Messages Notifications

Showing rows: 1 to 5

	rank bigint	customer_name character varying (100)	segment character varying (30)	region character varying (20)	total_sales numeric
1	1	Thais Sissman	Consumer	West	4.84
2	1	Thais Sissman	Consumer	South	4.84
3	3	Lela Donovan	Corporate	Central	5.30
4	4	Carl Jackson	Corporate	East	16.52
5	5	Mitch Gastineau	Corporate	South	16.74

- **Customers Who Made Only One Order** - Uses HAVING COUNT = 1 to find customers who placed only a single order — a churn risk segment.

Output -

Data Output Messages Notifications

Showing rows: 1 to 12

	customer_name character varying (100)	segment character varying (30)	region character varying (20)	order_count bigint	total_sales numeric
1	Jenna Caffey	Consumer	West	1	1058.11
2	Susan MacKendrick	Consumer	East	1	1043.05
3	Theresa Coyne	Corporate	East	1	1038.26
4	Jocasta Rupert	Consumer	South	1	863.88
5	Patricia Hirasaki	Home Office	South	1	729.65
6	Anthony O'Donnell	Corporate	West	1	161.28
7	Roland Murray	Consumer	East	1	98.35
8	Anemone Ratner	Consumer	South	1	88.15
9	Ricardo Emerson	Consumer	East	1	48.36
10	Mitch Gastineau	Corporate	South	1	16.74
11	Carl Jackson	Corporate	East	1	16.52
12	Lela Donovan	Corporate	Central	1	5.30

Total rows: 12	Query complete 00:00:00.458
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- **Customers With Above-Average Total Sales** - A CTE computes total sales per customer and filters those who exceed the average customer spend.

Output -

Data Output Messages Query Query History Notifications					
Showing rows: 1 to 1000					
	customer_name character varying (100)	segment character varying (30)	region character varying (20)	total_sales numeric	avg_sales numeric
1	Sean Miller	Home Office	South	25043.07	2896.85
2	Sean Miller	Home Office	West	25043.07	2896.85
3	Sean Miller	Home Office	Central	25043.07	2896.85
4	Sean Miller	Home Office	East	25043.07	2896.85
5	Tamara Chand	Corporate	West	19052.22	2896.85
6	Tamara Chand	Corporate	Central	19052.22	2896.85
7	Tamara Chand	Corporate	East	19052.22	2896.85
8	Tamara Chand	Corporate	South	19052.22	2896.85
9	Raymond Buch	Consumer	Central	15117.35	2896.85
10	Raymond Buch	Consumer	South	15117.35	2896.85
11	Raymond Buch	Consumer	East	15117.35	2896.85
12	Raymond Buch	Consumer	West	15117.35	2896.85
13	Tom Ashbrook	Home Office	Central	14595.62	2896.85
14	Tom Ashbrook	Home Office	East	14595.62	2896.85
15	Adrian Barton	Consumer	East	14473.57	2896.85
16	Adrian Barton	Consumer	West	14473.57	2896.85
17	Adrian Barton	Consumer	Central	14473.57	2896.85
18	Adrian Barton	Consumer	South	14473.57	2896.85
19	Ken Lonsdale	Consumer	East	14175.23	2896.85
20	Ken Lonsdale	Consumer	Central	14175.23	2896.85
Total rows: 1014		Query complete 00:00:00.164			

- **Highest Order Value per Customer** - A correlated subquery returns each customer's single largest order by sales value.

Output -

Data Output

Showing rows: 1 to 1000  						Page No: 1
	customer_name character varying (100)	segment character varying (30)	order_id character varying (25)	highest_order_value numeric	order_date character varying	
1	Sean Miller	Home Office	CA-2014-145317	22638.48	3/18/2014	
2	Sean Miller	Home Office	CA-2014-145317	22638.48	3/18/2014	
3	Sean Miller	Home Office	CA-2014-145317	22638.48	3/18/2014	
4	Sean Miller	Home Office	CA-2014-145317	22638.48	3/18/2014	
5	Tamara Chand	Corporate	CA-2016-118689	17499.95	10/2/2016	
6	Tamara Chand	Corporate	CA-2016-118689	17499.95	10/2/2016	
7	Tamara Chand	Corporate	CA-2016-118689	17499.95	10/2/2016	
8	Tamara Chand	Corporate	CA-2016-118689	17499.95	10/2/2016	
9	Raymond Buch	Consumer	CA-2017-140151	13999.96	3/23/2017	
10	Raymond Buch	Consumer	CA-2017-140151	13999.96	3/23/2017	
11	Raymond Buch	Consumer	CA-2017-140151	13999.96	3/23/2017	
12	Raymond Buch	Consumer	CA-2017-140151	13999.96	3/23/2017	
13	Tom Ashbrook	Home Office	CA-2017-127180	11199.97	10/22/2017	
14	Tom Ashbrook	Home Office	CA-2017-127180	11199.97	10/22/2017	
15	Hunter Lopez	Consumer	CA-2017-166709	10499.97	11/17/2017	
16	Hunter Lopez	Consumer	CA-2017-166709	10499.97	11/17/2017	
17	Hunter Lopez	Consumer	CA-2017-166709	10499.97	11/17/2017	
18	Adrian Barton	Consumer	CA-2016-117121	9892.74	12/17/2016	
19	Adrian Barton	Consumer	CA-2016-117121	9892.74	12/17/2016	
Total rows: 2508		Query complete 00:00:19.932				